Polynomial Fitting

We aim to fit a polynomial of order $N-1$ to N data points (t_i, y_i).

$$P(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_{N-1} t^{N-1}$$

The data points are generated using the following function:

$$y = \frac{1}{1+t^2}$$

where t is a vector of N equally spaced points in the interval $[-3, 3]$.

Problem

- Formulate the problem into a linear system $A\mathbf{x} = \mathbf{y}$, where $\mathbf{x} = [a_0, a_1, \ldots, a_{N-1}]^T$.
- Write a function to construct the matrix A and solve for the coefficient vector $\mathbf{x}$.

Implementation

Goal: Complete the function below.

```
import numpy as np

def fit_polynomial(N):
    t = ...  # Generate N points for t between -3 and 3
    y = ...
    A = ...
    x = ...

    return A, y, x
```

Hint: use `numpy.linspace` for the domain t.

Input

- An integer N representing the number of data points. $10 < N < 500$.

Constraints

- The degree of the polynomial must be exactly $N-1$.
- Use `numpy` for all matrix and vector operations.